

Modeling miRNA repressive pressure on the Hallmark programs of cancer

A conceptual and implemented framework. This document is the connective layer between the two reference docs: it explains *what we are modeling and why each design choice is made*, then ties every concept to its formula and its code. The two companions are:

- **FORMULAS.md** — the exact, code-faithful formula for every quantity (the *how*).
- **LANDSCAPE_REPORT.md** — the results: what the model found in TCGA-BRCA (the *what*).

Where this document gives a formula, the authoritative version is FORMULAS.md; where it gives a number, the authoritative version is LANDSCAPE_REPORT.md. The older METHODS.md / BIOLOGICAL_SYNTHESIS.md are consistent on mechanics but may trail on the latest counts — defer to the two above.

One-sentence thesis. A miRNA exerts *repressive pressure* on a gene; that pressure is a **potential** (how much repression the wiring + abundance *could* deliver) which may or may not be **realized** (an observed anti-correlation after confounders are removed); we model both, at five nested resolutions, across the healthy→tumor trajectory, and validate at the mRNA layer, the protein layer, and in independent cohorts.

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0. The modeling problem and the stance

0.1 What “miRNA pressure” is

A microRNA (miRNA) is a ~22-nt RNA that, loaded into the RISC/AGO complex, base-pairs a target mRNA's 3'UTR and represses it (mRNA destabilization predominantly, some translational block). One miRNA targets **many** genes; one gene is targeted by **many** miRNAs. The biological object we want to quantify is therefore not a single binding event but a **field of repression**: how much downward pressure the entire miRNA layer places on a gene, a program, a tumor.

We call this **pressure**. It is deliberately a *potential* quantity — built from **which edges exist** (curated/predicted interactions), **how strong each edge is** (evidence), and **how abundant the miRNA is** (expression). It does *not* by itself prove that repression happened in a given tumor. That second, harder question — *was the pressure realized?* — is answered separately by **coupling** (an observed, confounder-adjusted anti-correlation between the miRNA and its target).

Keeping potential and realization as two separate measurements, then joining them, is the central design decision of the whole framework. Everything else hangs off it.

0.2 Why Hallmarks, why breast cancer, and why it generalizes

- **The 50 MSigDB Hallmark gene sets** are a curated, non-redundant atlas of the coherent biological *programs* of cancer (proliferation, EMT, hypoxia, apoptosis, p53, interferon response, ...). Using them as the gene universe means every result is automatically interpretable as “pressure on a *named program*,” and lets us roll pressure up and down a biological hierarchy (§1).
- **Breast cancer (TCGA-BRCA)** is the implementation substrate because it has the deepest matched multi-omics (miRNA-seq, RNA-seq, CNV, methylation, ATAC, and two independent CP-TAC proteomes), plus well-defined molecular subtypes (PAM50) and independent replication cohorts (Buffa 2011).
- **The framework is substrate-agnostic.** Nothing in the pressure/coupling/ architecture machinery is breast-specific; only the *prior* (the tumor-direction polarity, §5.2) and some confounders (HRD) are cancer-flavoured. The same model runs on any cohort with miRNA + mRNA and any gene-set collection.

0.3 The stance: a few non-negotiable principles

These recur everywhere; they are stated once here and assumed thereafter.

1. **Static edge weight × dynamic abundance.** The edge weight $w(m, g)$ is fixed (it encodes curated/sequence evidence, which does not change between samples or states); all sample-to-sample and state-to-state variation enters through the miRNA’s **expression**. This cleanly separates “what the literature says is possible” from “what this tumor is actually expressing.”
2. **The prior is non-circular.** Edge weights, the HE filter, breast re-ranking, gene roles — none of these ever use TCGA expression. So they cannot leak into the coupling findings they are tested against. (This is why, e.g., the breast-context re-ranking in §2.6 is an *overlay*, never folded into the pressure budget.)
3. **Always report the adjusted and the raw, the gated and the ungated.** Every coupling has a raw Spearman beside its confounder-adjusted partial Spearman; every gated pressure has its ungated twin. Differences are read as *sensitivity*, not as the headline.
4. **Potential ≠ realization ≠ causation.** Pressure is potential; coupling is an observed association (still not binding); a luciferase/CLIP assay would be causation. The framework is honest about which rung a claim sits on, and tags every discovery S / P / H / R accordingly (§9).

1. The hierarchy of resolution

The single most important structural idea is that **the same pressure object is modeled at five nested resolutions**, and each resolution answers a different question with a different aggregation. Zooming in localizes a signal; zooming out contextualizes it.

UNIVERSE	all 50 Hallmark sets (4,384 genes; 1,410 under miRNA pressure), 721 arms, 5,108 edges
└─ HALLMARK	one program (e.g. EMT, P53_PATHWAY)
└─ TARGET-SET	the genes one miRNA arm / seed family represses ($T(m)$)
└─ GENE	one target gene g and its full crowd of regulators $R(g)$
└─ EDGE	one (miRNA $m \rightarrow$ gene g) interaction

Resolution	Unit	The question it answers	Pressure metric	Coupling metric
Edge ($m \rightarrow g$)	one interaction	Does <i>this</i> miRNA repress <i>this</i> gene?	$c(m, g, s)$ (§4.1)	per-edge partial- p (§6.3)
Gene g	one gene + its regulators $R(g)$	How much total pressure lands on the gene; who carries it?	$\text{pressure}(g, s) = \sum_m \text{gene_corepression } c$ (§4.1); gene-side share (§4.4)	(§6.3)
Target-set $T(m)$	one arm's / family's targets	Does the arm coordinately repress its whole module?	per-arm mass / share (§4.3)	module composite p (§6.3)
Hallmark	one program's genes	Net effect of the miRNA layer on a program's <i>output</i>	architecture program_output_change (§5)	per-Hallmark coupling count (§6.3)
Universe	all programs	Cross-program structure (subtype programs, NMF)	NMF factors, cross-set roll-ups (§5.6, LANDSCAPE §6)	fraction of programs coupled (§6.3)

Two facts make this hierarchy non-trivial (not just “sum the small things”):

- **Scope recomputation.** Every mean / SD / median / softmax that feeds pressure is recomputed *within the samples in the current scope* (whole cohort, or one PAM50 subtype, or one state). A subtype table is therefore **not** the cohort numbers re-sliced — the relative abundances are re-derived inside the subtype.
- **Different aggregations carry different meaning.** A gene's pressure is the **sum** over its regulators; a program's effect is a **topology-weighted, sign-aware propagation** over its genes (§5), not a sum. The roll-up rule is part of the model.

2. The edge universe

The edges are the skeleton: the set of (miRNA $\rightarrow$ gene) interactions over which pressure is built. The universe is **primarily experimental, with a controlled computational extension**.

2.1 Experimental core + computational extension

Layer	Source	What it contributes	Role
Experimental core	miRTarBase functional MTI (reporter assay, Western/protein, qPCR, perturbation, CLIP)	curated, literature-validated interactions	the spine — only these enter default (M0) pressure
Sequence prediction	TargetScan 7.1 (weighted context++ score)	site strength from seed match + conservation; study-count-independent	extension / orphan screens / hybrid modes (M7/M11)
Crosslink depth	ENCORI AGO-CLIP / degradome	physical-occupancy depth boost on existing pairs	depth boost (α) on ~25% of pairs

The default landscape spine is **miRTarBase-only (M0)**: 5,108 edges over 721 expressible arms and 1,410 target genes (cross-state landscape figures; LANDSCAPE_REPORT.md §0). Sequence-only (“orphan”) pairs do **not** enter the default budget — they are screened separately (§9.3) so that prediction noise can never inflate the headline. The hybrid modes (M7 = HE miRTar + capped TargetScan fill-in; M11 = pure TargetScan) exist as sensitivity layers, not the default.

2.2 The “high-evidence” (HE) gate

An edge is **high-evidence** if it has functional support from a *direct* assay:

```
HE(m,g) ⇐ n_functional_mti_studies ≥ 1
          AND ( n_reporter__functional_mti ≥ 1 OR n_protein__functional_mti ≥ 1 )
          (fallback: evidence_score ≥ 3 when cross-count columns unavailable)
```

The HE gate is a *binary membership filter* (which edges are credible enough to model at all); the edge **weight** below is the *continuous strength* of those that pass.

2.3 Edge weighting — component by component, with rationale

The edge weight has two factors: an **evidence numerator** (how well-supported the interaction is) divided by a **promiscuity denominator** (how thinly the miRNA spreads itself). Both are deliberate.

2.3a Evidence numerator — confidence_logclass Crucially, the weight does **not** use miRTarBase’s raw study-count field. That field inflates with publication count and treats a CLIP proximity hit like a luciferase proof. Instead the spine recomputes a **quality-weighted, log-damped** score from the per-assay-class Functional-MTI counts:

```
confidence_logclass(m,g)
= 3.0 · loglp(n_reporter__functional_mti)    ← luciferase: direct proof of repression
+ 2.5 · loglp(n_protein__functional_mti )    ← Western/protein: direct
+ 1.5 · loglp(n_rna__functional_mti )       ← qPCR/mRNA level
+ 1.5 · loglp(n_perturbation__functional_mti)
+ 0.5 · loglp(n_binding__functional_mti )    ← CLIP only: proximity ≠ repression
+ 0.3 · [same sum over the curator-flagged "weak" columns]
```

Three rationales are baked in:

- **Assay-directness weighting** (3.0 ... 0.5): a reporter assay that *demonstrates* repression earns 6× a binding-only CLIP hit that only shows *proximity*. This is the single most important correction — it stops physical-occupancy data from masquerading as functional proof.
- **loglp within each class**: diminishing returns. The 10th luciferase paper adds little over the 3rd; loglp defuses publication-count inflation so a heavily-studied textbook edge doesn’t crush a solid but less-famous one.
- **δ = 0.3 weak discount**: curator-flagged weak evidence still counts, but at 30%.

2.3b ENCORI depth boost For the ~25% of pairs with an ENCORI entry, physical AGO-CLIP depth is added at α=0.5:

```
enc_depth(m,g) = 2·loglp(clipExpNum) + 1·loglp(degraExpNum)
                + 0.5·I(TargetScan ∧ miRanda ∧ PITA) + 0.25·loglp(pancancerNum)
```

```
evidence_score_eff(m,g) = confidence_logclass(m,g) + 0.5 · enc_depth(m,g)  (0 if no ENCORI row)
```

The base edge strength is then $w_0(m,g) = \log(1 + \text{evidence_score_eff})$.

2.3c Promiscuity denominator — the “one miRNA, many genes” problem A hub miRNA that targets *everything* should not be allowed to dominate any one gene merely by being promiscuous. So w_0 is divided by a per-arm **mass denominator** (default evidence_mass):

$D(m) = \log(1 + \sum_{g'} \log(1 + \text{evidence}(m, g')))$ ← total curated target mass of arm m
 $\text{edge_w}(m, g) = w_0(m, g) / \max(D(m), 1e-9)$

Mode	$D(m)$	Reading
none	1	raw evidence (no promiscuity control)
degree	$\log(1 + \#\text{targets}(m))$	penalize <i>number</i> of targets
evidence_mass (default)	$\log(1 + \sum \log(1+ev))$	penalize <i>total curated mass</i>
combined_mass	adds the TargetScan-mass term	miRTar + TS (hybrid modes)

Rationale (measured, held_out_tuning + denominator_attribution_sweep): $D(m)$ is a single per-arm scalar, so it acts on attribution, **not** coupling — and we verified this rather than assert a coupling benefit. (i) **Coupling-neutral:** in 5-fold patient CV the denominator barely moves gene-level coupling — bare none is in fact marginally *ahead* out-of-sample and the spine sits within ~ 0.001 of it; the candidate ranking reproduces perfectly across folds (train→test rank- p 1.0), so the choice is **not overfit** (non-circular) but is also **not coupling-maximal**. (ii) **Specificity-invariant:** spec is a within-arm ratio in which $D(m)$ cancels exactly (rank- p 1.0). (iii) **Where it acts — cross-arm attribution:** it re-assigns a gene’s dominant regulator for $\sim 11\%$ of genes (dominant-arm concordance 0.89 vs none) and de-concentrates the hub structure (Gini 0.570→0.538; dominance spread from ~ 43 to ~ 56 arms) — exactly the promiscuity control it is for. **evidence_mass** penalizes by *credible* evidence mass rather than raw degree (which de-concentrates marginally more) — a principled middle, not a uniquely optimal pick. **Note:** the old “sharpest/most stable coupling (Basal $\geq 42/50$)” rationale was a coupling-axis claim that the held-out refutes; the spine’s value is on attribution.

2.4 Many-to-many, made explicit

The two directions of the many-to-many graph are handled by **two different mechanisms**, and conflating them is the classic mistake.

- **One miRNA → many genes** is handled by the **denominator** $D(m)$ (§2.3c) and by **specificity** (§4.4). The denominator stops a hub from dominating; specificity measures how concentrated an arm’s budget is on a single gene ($\text{spec}(m, g)$ high = focused/clean readout; low = promiscuous hub).
- **Many miRNAs → one gene** is handled by the **gene-level aggregation** ($\text{pressure}(g) = \sum_m c$, §4.1) and by **gene-side share** (gene_share , §4.4) — what fraction of a gene’s *total incoming pressure* one arm carries, with a rank (1 = the gene’s dominant regulator). A gene like PTEN has ~ 91 regulators; gene-side share is how we say “miR-9-5p is PTEN’s #1 regulator” while that edge is a trivial slice of miR-9’s own portfolio (arm-side edge_share 0.006). **edge_share is arm-side, gene_share is gene-side — they answer opposite questions** and both are kept.

2.5 Membership is many-to-many too (gene $\in N$ Hallmarks)

A gene that belongs to N Hallmark sets generates N (miRNA, gene, hallmark) edge rows. Per-Hallmark views keep all N ; cohort-level counts **dedup on (miRNA, gene)** so a multi-program gene is not counted N times.

2.6 The pan-context limitation and the non-circular priors (overlay, not budget)

miRTarBase carries **no tissue field** — an edge is “high-evidence” if validated in *any* cell line. Only **12.3%** of HE edges have *any* breast-specific validating paper. This is a real limitation of the prior, and it is quantified rather than hidden:

- **Breast re-ranking** ($\text{edge_breast_context}$): mine each edge’s Functional-MTI PMIDs for breast MeSH / mammary cell lines, re-rank a gene’s regulators by $\log_{1p}(\text{evidence}) + \beta \cdot \log_{1p}(n_{\text{breast_pmids}})$. This flips the *credited top regulator* for $\sim 90/783$ multi-regulator

genes and recovers textbook breast edges (PTEN→miR-21-5p, ERBB2→miR-125b-5p). **It never touches the pressure budget** — it is an audit/nomination overlay, kept strictly non-circular (literature only, no TCGA expression).

- **Multi-axis prior** (edge_prior_refinement): composes evidence with four more auditable, TCGA-independent axes — TargetScan sequence strength, breast literature, *negative* evidence (Non-Functional MTI contradictions; ~26% of HE edges are contradicted ≥once), and miR-GeneDB guide/passenger status — each a **separate visible term** so nothing is hidden. Also an overlay, not the budget.

The lesson encoded here: **the spine prior is pan-context and we say so loudly; the breast-specific refinements are kept separate precisely so they can be tested rather than assumed.**

3. Expression modeling — the dynamic piece

The edge weight is static; *all* sample/state variation enters through expression. There are four building blocks, combined into modes. (FORMULAS.md §3-4 is canonical.)

Notation: $x[m, s] = \log_2(\text{RPM}+1)$ of arm m in sample s ; mean_m , sd_m , med_m are in-scope statistics; $R(g)$ = the arms regulating g .

3.1 The four building blocks

- (a) z-score (dynamics) $z[m, s] = (x[m, s] - \text{mean}_m) / \text{sd}_m$
- (b) softmax share $\text{sm}[m, s] = \exp(x[m, s] - \text{med}_m) / \sum_{m' \in R(g)} \exp(x[m', s] - \text{med}_{m'})$
- (c) absolute anchors $\text{logrpm}[m, s] = \max(x[m, s], 0)$
- (d) healthy-anchored dev $\text{dev}[m, s] = x[m, s] - \text{median_healthy}(m)$ (log2 units, no SD)

- **z (a)** — deviation of the arm from *its own* typical level across the cohort. Signed, unitless. *Caveat:* an arm sitting at its mean has $z \approx 0$ and contributes nothing — a stably-high arm is invisible under z . This is intentional for coupling (where *dynamics* are the signal) but wrong for attribution (§4.3).
- **softmax share (b)** — the fraction of gene g 's regulator “abundance budget” held by arm m in sample s ; positive, sums to 1 across $R(g)$. This is where *the crowd of regulators on one gene* enters: an arm only gets share to the extent it out-abounds its co-regulators on that gene.
- **absolute anchor (c)** — keeps absolute level in the score so a relative share between two barely-expressed arms isn't treated like one between two highly-expressed arms.
- **healthy dev (d)** — replaces the within-tumor z with deviation from the **healthy (GTEx-breast) median**. No SD division, so a uniformly tumor-elevated oncomiR is *not* gated to ~ 0 the way z would gate it. This is the engine of the acquired axis (§8).

3.2 The three headline modes (and why each exists)

Mode	Formula	Sign	Used for	Why
softmax_z_logrpm	$\text{sm} \cdot z \cdot \text{logrpm}$	$\pm$	coupling spine	relative share $\times$ within-tumor dynamics $\times$ absolute level; z centres on the tumor-cohort mean $\rightarrow$ answers “ <i>who moves across tumors</i> ”

Mode	Formula	Sign	Used for	Why
softmax_logrpm	$sm \cdot \logrpm$	+	attribution / shares	no z, so nothing is zeroed for being stably expressed; $abs_share == signed_share$; cross-state comparable
softmax_devhealthylogrpm	$\frac{m \cdot \logrpm}{\logrpm}$	$\pm$	acquired axis	centres on the <i>healthy</i> median → “who carries pressure the tumor acquired over healthy tissue”

The reason there are three (not one) is that they answer three *different* questions, and using the wrong one silently corrupts the answer — z for coupling (dynamics), no-z for attribution (who carries the steady-state pressure), healthy-dev for acquisition. This is the “Q1 attribution policy” (FORMULAS.md §“Attribution policy”). **Note:** these modes build the *aggregate* (gene/program) pressure; a single-edge or single-module coupling does **not** apply the transform — it uses the arm’s raw abundance directly (§6.3).

4. Pressure — the master formula and its variants

4.1 The one master formula

Everything in §4-5 is a piece of, or a summary of, this:

$$\text{pressure}(g, s) = \underbrace{\sum_{m \in R(g)}}_{\substack{\text{aggregate} \\ (\text{sum over} \\ \text{the gene's} \\ \text{regulators})}} \cdot \underbrace{\text{expr_mult}(m, s)}_{\substack{\text{expression} \\ \text{piece (§3)}}} \cdot \underbrace{\text{edge_w}(m, g)}_{\substack{\text{evidence} \\ \text{piece (§2)}}$$

per-arm contribution: $c(m, g, s) = \text{expr_mult}(m, s) \cdot \text{edge_w}(m, g)$

$c(m, g, s)$ is the **edge-resolution pressure** — the atom. The gene-resolution pressure is its sum over $R(g)$. **Every share, specificity, architecture, and family number is built from c.** Default aggregate = **sum** (a gene with more, stronger regulators genuinely has more pressure on it); mean is a sensitivity variant.

4.2 Acquired / lost pressure — at every resolution

“Acquired” pressure is pressure the tumor *gained over the healthy state*; “lost” is pressure it shed. It is defined identically at each resolution by differencing the same c across states $X \in \{\text{gtex}, \text{nat}, \text{tumor}\}$ (cross-state uses the no-z softmax_logrpm, because z is standardized *within* a state and is not comparable across states):

```
EDGE   gain_gtex(m,g) = c_tumor(m,g) - c_gtex(m,g)           (the edge's acquired increment)
GENE   acquired_vs_gtex(g) = pressure_tumor(g) - pressure_gtex(g) (principal healthy anchor)
       acquired_vs_nat(g)  = pressure_tumor(g) - pressure_nat(g)  (collapse-free same-platform cross-check)
HALLMARK pro_tumor_acquired(m,set) = prior(set) ·  $\sum_g [-\max(c\_tumor - c\_gtex, 0)] \cdot \text{output\_influence}$ 
```

The two healthy anchors are complementary: **GTEx→tumor** captures the large healthy-baseline jump (the miR-21 program, top acquired genes PCBP1/PYCR1/PTPN14/...); **NAT→tumor** is on the *same platform* (no cross-platform artifact) and ranks the final malignant step (HUS1/ENG/PTEN/INHBA). See §8 for the full trajectory machinery, the two acquired axes (rank vs magnitude), and the GTEx collapse-repair.

4.3 Share — who carries a gene’s pressure

Share decomposes a gene’s total pressure among its regulators. There are two axes of choice (global vs per-sample, magnitude vs signed):

```
global_abs_share(m) = mean_s |c(m,:) | / Σ_{m'} mean_s |c(m',: ) | ← PRIMARY RANKING
global_signed_share(m) = mean_s c(m,:) / Σ_{m'} mean_s c(m',: ) ← COHERENCE DIAGNOSTIC
mean_abs_share(m) = mean_s [ |c(m,s) | / Σ_{m'} |c(m',s) | ]
mean_pos_share(m) = mean_s [ max(c,0) / Σ_{m'} max(c',0) ]
```

- **Rank arms on global_abs_share** — a single stable ratio (magnitude share of the gene’s total pressure).
- **Diagnose with global_signed_share** — when it diverges from the abs version, arms are *cancelling* (some net-activating in a sample). Can be <0 or >1; diagnostic only.
- Under the **no-z attribution mode** every $c \geq 0$, so abs == signed and the two collapse — which is the whole point of attributing under softmax_logrpm.

share exists in **two complementary directions** (§2.4): arm-side edge_share (this gene’s fraction of the arm’s own budget) and gene-side gene_share (this arm’s fraction of the gene’s incoming pressure, with a rank).

4.4 Specificity — focused target vs promiscuous hub

How concentrated is an arm’s budget on one gene? (Two flavours, because orphans have no curated edge to measure realized share on.)

```
realized: spec(m,g) = mean_s |c(m,g,:) | / Σ_{g' ∈ targets(m)} mean_s |c(m,g',: ) |
TargetScan: ts_spec(m,g) = ts_weight(m,g) / Σ_{g'} ts_weight(m,g')
```

spec is the fraction of the arm’s *total* target-mass on one gene, read against the even-spread baseline $1/|targets(m)|$ (well under 1% for a promiscuous arm). So $spec \geq 0.05 \rightarrow$ a disproportionately large, several-fold-above-baseline share = **specific/focused** = cleaner single-gene validation (a *relative concentration*, not an absolute majority); $spec \leq 0.01 \rightarrow$ at/below the floor = **promiscuous hub**. Specificity feeds the edge classification (driver vs minor vs structural-only) and the orphan validation priority (§9.3) — a focused edge is a cleaner wet-lab readout.

Seed-family non-identifiability applies to every per-arm number here (share, gene-side rank, spec). Paralogues sharing a seed (miR-29a/b/c, miR-302/372/373/520, ...) hit the same sites and co-vary, so a single-arm attribution may simply credit one member as a stand-in for its family. This is measured, not assumed: within-seed-family arm pairs co-vary $\approx 2\times$ random pairs (Mann-Whitney $p \approx 4e-14$) and the largest multi-arm hubs are 20–37% redundant (seed_family_justification). Read share/spec at the **seed-family** level as the safe unit; the specific within-family arm is a nomination. The realized-coupling layer uses the same seed-family grouping for its dependence-aware FDR and permutation null (§6.x, coupling_inference / coupling_permutation). Singleton-seed arms are unaffected.

5. Architecture — modeling a set of genes

This is the answer to “*how do we model a set of genes and their components*” — the hardest part conceptually, because a program is **not** a bag of independent genes. A miRNA repressing an upstream

activator changes the whole downstream program; repressing a tumor-suppressor is pro-tumor regardless of network position. The architecture layer (geneset_architecture) overlays three orthogonal lenses on otherwise-flat pressure. (FORMULAS.md §12; LANDSCAPE_REPORT.md §7.)

5.1 Lens 1 — topology (network position inside the program)

Build the program's regulatory wiring from **OmniPath** (directed + signed), induced on the set's members. Each ordered gene pair gets a **calibrated continuous sign** instead of a hard $\pm 1/0$:

$$\text{net_sign}(s \rightarrow t) = (n_{\text{stimulation}} - n_{\text{inhibition}}) / (n_{\text{stimulation}} + n_{\text{inhibition}}) \in [-1, +1]$$

(Pure activator $\rightarrow +1$, pure inhibitor $\rightarrow -1$, conflicting $\rightarrow$ a graded value — instead of flattening dual edges to 0, which demonstrably over-counted.) Then a finite-horizon signed-path propagation (Katz, $\alpha=0.25$, 6 hops) gives each gene its **net downstream influence**:

$$T = \sum_{k=1..6} (\alpha \cdot S)^k \quad \text{output_influence}(g) = 1 + \sum_j T[g, j]$$

A central activator scores >0 ; an inhibitor hub <0 . Because a miRNA *represses* g , the effect on program output flips sign with the node's influence:

$$\text{program_output_change}(m) = \sum_g [-c_{\text{tumor}}(m, g) \cdot \text{output_influence}(g)]$$

>0 = the arm's repression nets to *activating* the program (it mostly hit inhibitor hubs); <0 = it *damages* the program. **This is the only lens that captures indirect, network-mediated effects** — everything else (flat pressure, coupling, gene-role) is local.

5.2 Lens 2 — tumor-direction prior (is the program pro- or anti-tumor?)

A coarse, hand-curated BC-context polarity per Hallmark (TUMOR_DIRECTION_PRIOR): +1 pro-tumor engines (EMT, HYPOXIA, GLYCOLYSIS, E2F, G2M, MYC, ...), -1 tumor-suppressive / anti-tumor-immune (APOPTOSIS, P53, IFN α/γ , INFLAMMATORY, ALLOGRAFT), 0 ambiguous (kept directional but not polarized).

$$\text{pro_tumor_pressure}(m, \text{set}) = \text{prior}(\text{set}) \cdot \text{program_output_change}(m)$$

>0 = pushes toward malignancy (activating a +1 engine, or **releasing a brake** on a -1 program). The headline mechanistic finding falls straight out: because miRNAs *repress*, they **damage** the +1 proliferation engines (negative pro-tumor on E2F/G2M/EMT) and turn malignant mainly by **brake-release on the -1 suppressive programs** (miR-21-5p $\rightarrow$ P53/APOPTOSIS, miR-375 $\rightarrow$ ALLOGRAFT/immune-evasion). All 50 sets are net *damaged* by the miRNA layer; only the *targeted sub-pattern* is oncogenic.

5.3 Lens 3 — gene-role / malignancy (driver-level direction)

Topology has a blind spot for canonical oncomiRs (it scores miR-21-5p ~neutral because it mostly damages the engines it broadly hits, and PTEN acts on a *lipid*, not a gene node). The fix is a curated **gene-role** layer (gene_roles.py: 232 high-confidence drivers — 114 oncogene / 106 TSG / 12 dual, COSMIC-CGC/OncoKB consensus) giving each gene malignancy_sign $\in \{-1 \text{ TSG}, 0, +1 \text{ onco}\}$:

$$\text{mal_pro_tumor}(m) = \sum_g [-\text{malignancy_sign}(g) \cdot c_{\text{tumor}}(m, g)] \quad (\text{graph-free driver call})$$

$$\text{mal_pro_tumor_arch}(m) = \sum_g [-\text{malignancy_sign}(g) \cdot c_{\text{tumor}}(m, g) \cdot w_{\text{arch}}(g)] \quad (\times \text{centrality MAG})$$

(w_{arch} = reverse-PageRank $\div$ redundancy, always ≥ 0 , so it scales magnitude without re-flipping the sign — an earlier version that multiplied by *signed* influence double-counted direction and was rejected.) This **recovers the textbook oncomiR roster** (miR-21/182/10b/93/106b/25/221) that topology missed — external validation that the layer captures real driver biology.

Topology and gene-role are kept side by side, not merged. They answer different questions (“what the wiring says” vs “what the driver annotation says”), and miR-21 is the canonical case where you must read both. **Rule: for high-breadth abundant arms ($n_{\text{targets}} \gg \text{median}$) read the gene-role / coupling / acquired axes, not the topology sign.**

5.4 The continuous, all-gene complement (DepMap dependency)

The binary malignancy_sign covers only 232 drivers. A graded, all-gene counterpart uses **DepMap 26Q1 CRISPR (Chronos) gene-effect over the 96-line breast panel**:

$\text{dep_role_weight}(g) = \text{gene_effect_breast}(g)$ (<0 = essential/dependency $\rightarrow$ repress = anti-tumor;
 ≥ 0 = TSG-like $\rightarrow$ repress = pro-tumor)
 $\text{mal_pro_tumor_cont}(m) = \sum_g \text{dep_role_weight}(g) \cdot c_tumor(m,g)$ (ALL genes, 530/543 arms scored)

CRISPR resolves the *dependency* (anti-tumor) tail well but the TSG tail weakly, so this is the **all-gene graded onco-collateral / anti-tumor axis**; the curated *tsg_credit* stays the better *pro-tumor* detector — complementary (corr ≈ 0.29). It shows the miRNA layer lands hardest on proliferation dependencies (G2M, E2F, MYC).

5.5 The within-program functional role (TF hierarchy)

Topology centrality misses curated **transcription-factor** identity (only $\sim 10\%$ of hubs are curated TFs; $\sim 48\%$ of curated TFs are below-median centrality). A Lambert-2018 TF census adds *is_tf* / *hub_tf* (curated TF $\wedge$ topology hub = high-confidence master regulator) and a tunable hierarchy weight $w_hier = w_arch \cdot (1 + 0.5 \cdot is_tf)$, wired into TF-up-weighted versions of the malignancy and topology-damage scores. This sharpens the “which master regulators are hit” read: P53 $\rightarrow$ TP53, G2M $\rightarrow$ E2F1/2/3/MYC/MYBL2, IFN γ $\rightarrow$ STAT1.

5.6 Roll-ups

Each architecture score rolls up across the universe: per-arm cross-set (*sum_pro_tumor_pressure*, *n_sets_pro_tumor*, *topset*), per-set totals (*total_program_output_change*, *total_mal_pro_tumor*), seed-family aggregates, and an **acquired-weighted parallel** of every score ($\text{weight} = \max(c_tumor - c_gtex, 0)$) that isolates the *tumor-acquired* pro-tumor pressure from the constitutive (e.g. miR-21’s TSG repression is 93% acquired; miR-374a’s is constitutive). The universe-level unsupervised structure (NMF programs, subtype-divergent factors, within-pathway sub-modules) is the subject of LAND-SCAPE_REPORT.md §6 and is summarized in the glossary.

6. Coupling — realized pressure and confounding

Pressure is potential. **Coupling** asks whether the repression is actually *observed* in the cohort: does a tumor with more of the miRNA have less of the target, **after the obvious confounders are removed**?

6.1 The core idea

Realized repression should show up as a **negative correlation** between the miRNA *dose* and target expression — the dose being the arm’s raw abundance at single-regulator resolution (edge/module) and the aggregate pressure where many regulators converge (gene/program); see §6.3. But a naive correlation is dominated by confounders (a tumor that is more cellular, more aneuploid, more proliferative drags *everything* with it). So we use a **partial Spearman**: regress the confounders out of *both* the miRNA and the target, then correlate the residual ranks. A negative partial- ρ that survives = realized repression. Spearman (rank) rather than Pearson because expression is heavy-tailed and we want monotone, not linear, association.

6.2 The confounders — the deep dive

This is where most of the model’s defensiveness lives. Each canonical confounder is included for a *specific* mechanistic reason, and each is mechanistically motivated, not just thrown in.

Confounder	What it is	Why it confounds miRNA↔target	Source
Purity (CPE)	Consensus tumor-purity estimate	Stroma/immune infiltrate carries its <i>own</i> miRNAs and mRNAs; a low-purity tumor looks “high miR / low target” for purely compositional reasons	analysis.utils.common.loaders
HRD	Homologous-recombination deficiency / genomic scar	Genome-instability burden co-drives both arms; HRD-high (basal) tumors over-express many loci	clinical
Target copy number	ASCAT3 integer CN of the <i>target</i> gene	If the target is amplified, its mRNA rises for a reason that has nothing to do with the miRNA — masking or faking repression	load_cnv_target_genes
Proliferation	E2F+G2M metagene (+ two alternatives)	Cell-cycle drags both miRNA clusters (miR-17~92) and their targets; <i>must</i> be removed to avoid “cell-cycle in disguise”	metagene
ESTIMATE (+epithelial)	immune + stromal (+ epithelial + prolif) composite — estimate_epi, the default	The most complete composition control; the epithelial axis specifically de-confounds lineage co-expression (miR-200 family)	ESTIMATE ssGSEA
Sequencing batch	TCGA analyte-plate dummies — miRNA-seq + RNA-seq plates (plate_both; rare plates pooled → ~14 levels/assay); per-state NAT/GTE _x batch (barcode plate / GTE _x SMNABTCH·SMGEBTCH) in cross-state; TMT-plex in CPTAC	Technical run/plate co-varies features on <i>both</i> arms, so it can fabricate or mask coupling; adjusted as covariate dummies — not ComBat (ComBat over-corrects the unbalanced TCGA design and can manufacture the feature-feature correlations that <i>are</i> the readout)	data_loaders.augment_tcga_batch / tcga_batch

Three subtleties make the proliferation handling careful rather than reflexive:

- **Three proliferation proxies**, differing in their E2F/MYC dependence: the full E2F+G2M metagene, an **E2F/MYC-stripped** G2M+spindle metagene, and **MKI67 alone**. Reporting all three tests whether repression survives proliferation adjustment *without over-correcting through a shared E2F path* — the “we just removed the signal” objection.
- **Proliferation is a suppressor confound, not a driver of the signal**. Within Basal it is positively associated with *both* pressure and target expression, so adjusting for it makes the

negative coupling *more* negative (it was masking, not faking). This is why the basal coupling *strengthens* under proliferation adjustment — a key robustness result, not a fragility.

- **The covariate ladder.** Coupling is reported as a *nested ladder* of covariate sets, not a single number, so a reader sees exactly what each adjustment costs:

```
raw_rho           none
rho_CPE_HRD       CPE, HRD
rho_CPE_HRD_CN    + target CN
rho_e2f_g2m       CPE, HRD, E2F+G2M
rho_e2f_g2m_CN    + target CN          ← "proliferation-surviving" headline
rho_mki67(_CN)    MKI67-only proxy
rho_ortho_noE2F_MYC(_CN) E2F/MYC-stripped proxy
```

Since the **2026-06-26 Normal-excluded + batch refresh, TCGA sequencing-batch dummies are appended to every non-raw rung** (augment_tcga_batch), so each confounder-adjusted ρ above is *also* batch-corrected (the raw rung is the only batch-free one); cross-state coupling adds the per-state NAT/GTex batch at the same chokepoint, and CPTAC adds its TMT-plex.

Significance flags, **BH-FDR within each (predictor_type, predictor, scope) family** (this “family” is the *statistical testing family* — the set of tests sharing a correction; distinct from the *miRNA seed family* below):

```
neg_sig_cn_fdr      = (rho_CPE_HRD_CN < 0) AND (q_CPE_HRD_CN < 0.05)      ← headline
neg_sig_prolif_cn_fdr = (rho_e2f_g2m_CN < 0) AND (q_e2f_g2m_CN < 0.05)  ← proliferation-surviving
```

A **variability gate** ($sd_x < 0.5$ AND $IQR_x < 0.75$ on log2RPM) marks flat arms whose “null coupling” should not be over-read.

Two senses of “family”, both handled (coupling_inference / coupling_permutation):

- *Statistical testing family* — the BH scope above. Made dependence-robust with **Benjamini-Yekutieli** (q_{by}), since correlated tests violate BH’s independence.
- *miRNA seed family* — paralogues sharing a seed are non-independent (justified empirically: within-family arm $\rho \approx 2\times$ between, multi-arm hubs 20–37% redundant, seed_family_justification). Where the test unit is a miRNA (edge / target-set), each seed family is collapsed to one **Simes** test ($q_{seedfamily}$) plus a min-statistic family-wise null ($p_{seedfamily_fwer}$); the de-duplicated discovery count is in *families*, not edges.

Permutation null at every rung. Beyond the asymptotic partial- ρ p , every coupling is re-tested by **Freedman-Lane** rank-residual permutation, applied jointly (one shared sample permutation per draw) so seed-family dependence is preserved in the null; a smooth tail- p from the null moments feeds the FDR (so the empirical $1/(B+1)$ floor never caps it) while the discrete exceedance p is kept as the honest check. The same joint null gives the universe “how many programs couple” count its own null. Resolutions: edge, gene, target-set, program, universe.

6.3 Coupling at every resolution

The partial- ρ machinery is applied at each rung of the hierarchy (§1), against a different predictor/target pair:

Resolution	Predictor	Target	Coupling object	Module
Edge	arm $x[m, :]$	target $y[g, :]$	per-edge partial- ρ , per state; the coupling_sig_pattern (R/P/O over GTex/NAT/tumor)	mirna_state_class, edge_partial_corr

Resolution	Predictor	Target	Coupling object	Module
Gene	aggregate incoming pressure Pagg(g)	gene's own expression y[g, :]	gene_corepression is the <i>gene</i> net-repressed? (282/1,424 in tumor — GATA3, ESR1, AURKB)	decoupling_validation §14a
Target-set	arm abundance x[m, :]	mean within-state z over T(m) (the module composite)	does the arm's <i>whole module</i> move down together? (miR-18a/106b/17~92)	mirna_comovement §13a
Family	family predictor (mean / max / sum / mean_z over the seed family)	target	family-as-unit coupling	mir301_family_depth §8
Hallmark	per-Hallmark pressure	per-Hallmark mean target expression	count of programs with FDR-neg coupling (Basal 42/50)	hallmark_interaction
Universe	all programs	—	fraction of programs/edges coupled; cross-cohort concordance	roll-ups

The **module composite** deserves emphasis as the strongest confidence-independent signal: for each arm, z-standardize each target within state, take the (optionally edge-weighted) mean over targets, then one partial-p of arm abundance vs that composite. A strongly negative composite-p means the *coordinated* module is repressed — something no single-edge p can see. (Edge-weighted $\approx$ equal-weight, $r=0.97-0.99$, so equal-weight is the default.) A **role-stratified** version (TSG vs onco-gene targets) is an interpretive *overlay*, deliberately not folded into one number — folding role into the composite would re-conflate “is it repressed” with “is it pro/anti-tumor,” which is the architecture layer’s job (§5).

Predictor switches with resolution — and only the share matters. The edge and aggregate predictors are not a reparametrization, but most of the difference is inert: the static edge weight is a sample-constant scalar (rank-invisible) and z is affine in abundance (rank-identical for one arm), so neither moves a single-edge coupling; the **softmax share** (which pulls in co-regulators) is the sole differentiator. A head-to-head check — the same CPE+HRD+target-CN partial-p under every predictor (coupling_predictor_comparison) — confirms it: at the **edge** level z reproduces abundance (sign-concordance 0.96, recovers 94% of FDR-sig edges) while share-alone recovers only ~49% and flips its median positive → **raw abundance is the most sensitive single-edge predictor** (why edge/module coupling uses it). At the **gene** level share-alone is degenerate (per-gene shares sum to ≈ 1 each sample → near-constant), but the full pressure recovers ~89% of the naive abundance-sum’s hits and finds **~2-3% more** with a deeper median → the construction folds many regulators into one dose better than summing abundances. Net: pressure differs from abundance only through the share, which *weakens* one edge but *strengthens* the gene aggregate (MH-44).

6.4 The potential × realized join (pressure × coupling; the headline classes)

The whole point of measuring *both* potential and realization is to **join them**. Here *pressure* is the **potential** axis (its edge-level trajectory tracks the miRNA’s abundance, the weight being static) and *coupling* is the **realized** partial-p — the join is potential × realized, **not** a correlation of one against

the other. Every edge carries a pressure (potential) trajectory (does the miRNA's abundance gain/lose over states) and a coupling (realized) R/P/O trajectory. Their join is the headline class:

acquired_realized	pressure rises AND coupling turns repressive (00R / P0R) ← genuinely sw.
field_established_realized	arm acquired + 0RR (repressive in NAT, kept in tumor)
acquired_unrealized	pressure up but coupling never realized
lost	pressure flat + brake released (R00: repressive in healthy, gone in tumor)
constitutive	repressive in both healthy and tumor (R0R)
nat_decoupled	NAT-only sign-flip vs both states (essentially absent as an FDR class)

This join is what lets us say, e.g., “*healthy breast actively holds ERBB2/CXCR4/MMP2/ NOTCH2 under miRNA repression and the tumor releases the brake*” (the R00 released-brake biology) as a **coupling-derived** claim, immune to the pressure-side caveats. See §8 for how the state axis is built and LAND-SCAPE_REPORT.md §3–4 for the full class biology and the de-coupling validation (CN/methylation/ATAC conditioning).

7. Protein realization — beyond the mRNA layer

miRNA repression ultimately acts on the **protein** product. The mRNA-level coupling (§6) sees destabilization; it cannot see pure translational block. So the framework tests whether pressure realizes onto **protein**, two ways, using CPTAC mass-spec proteomes.

7.1 Three layers

For a target gene with mRNA `rna_z` and protein `protein_z`:

L2 (direct protein)	partial- ρ (pressure , protein_z)	expect < 0
L1 (slope-1 RNA→protein)	partial- ρ (pressure , rna_z - protein_z)	expect > 0 (trans)
L1b (protein_resid)	partial- ρ (pressure , protein residualized on its own mRNA)	expect < 0 = mRNA effect

L1b is the key idea — “realization onto protein *through* vs *beyond* the mRNA.” Residualizing protein on its own mRNA removes everything the *measured* mRNA explains; a surviving negative ρ is a protein-level effect the bulk steady-state mRNA does not capture — best read as *beyond-(measured)-mRNA*, **not** strictly *translational*. It is consistent with a translational block but also with destabilization bulk RNA-seq misses (deadenylation / poly(A) shortening; 3'-UTR-isoform / APA effects), protein-vs-mRNA turnover kinetics (protein integrates past mRNA), or indirect routes — plus platform noise between sequencing and MS. This is the “with or without the miRNA-mRNA coupling” distinction: L2 reads protein *together with* whatever happened at the mRNA layer; L1b reads protein *after subtracting* the measured mRNA.

7.2 The two CPTAC cohorts and the headline

- **CPTAC TCGA-105** (Mertins, iTRAQ): the *same* TCGA patients re-assayed. Protein coupling is essentially null (0/1006 genes; L1b 0) — the noisier same-patient iTRAQ cannot see it.
- **CPTAC prospective-122** (Krug, TMT): *independent* patients, with its own GDC miRNA-seq, so pressure is rebuilt **self-contained** (no cross-cohort matching). Here the spine **validates at protein**: 34/1143 genes FDR-neg on L2 (led by **ZEB1**, the canonical miR-200 target, $\rho = -0.55$), 6/50 Hallmarks (EMT top).

The mechanistic conclusion: **the repression is predominantly mRNA-mediated**. Residualizing on mRNA collapses most of it (ZEB1 $-0.55 \rightarrow -0.24$ NS), leaving only ~8 genes with a genuine protein-beyond-mRNA (L1b) component — canonical mammalian miRNA biology (destabilization dominates, small translational residual). The L1b survivors are **known, subtype-structured edges**, and critically they **recover the basal miR-17~92→ PTEN hub at the protein layer** (miR-17-5p $\rho = -0.34$, $q = 0.008$) — the strongest external corroboration of the subproject's central hub thesis.

7.3 Target specificity (the decoy control)

A reviewer worry: maybe the survivor arms are just *globally* anti-correlated with the whole proteome (a confound). The **decoy control** (cptac_target_specificity) compares each driver arm's coupling to its *cognate* protein against its coupling to every protein it does **not** target (≥ 50 decoys). The cognate couplings (median $\rho = -0.33$) sit $\sim 16\times$ more negative than the decoy null (≈ -0.02 , centred at zero): 9/13 FDR drivers are target-specific (empirical $p < 0.05$), and the 2 borderline cases are exactly the broad proliferation oncomiRs (PTEN $\leftarrow$ miR-17-5p) whose decoy nulls are somewhat negative — i.e. the control correctly discounts genuinely-promiscuous arms. The specificity survives the literal TMT-plex batch adjustment.

8. The healthy $\rightarrow$ NAT $\rightarrow$ tumor trajectory (acquired pressure)

The “acquired” concept (§4.2) is given its own machinery (mirna_state_class) because *how* a miRNA's pressure moves across the three tissue states is itself a finding.

8.1 The three states and the healthy anchor

GTEX-healthy breast $\rightarrow$ TCGA-NAT (normal-adjacent) $\rightarrow$ TCGA-tumor
| | |
true healthy field-effect / pre-malignant malignant

The **healthy anchor** is GTEX breast (346 samples), **quantile-normalized onto the TCGA scale** and joined by **MIMAT accession** (not name). GTEX's uniquely-mappable pipeline *collapses* ~ 278 canonical arms to zero (let-7a, miR-30a, miR-182...); two imputation layers repair coverage:

1. **miTED anchor rank-transfer** (primary): arms measured in both GTEX and DIANA-miTED define a monotone map; a gap arm's GTEX-equivalent is predicted from its miTED value. Validated leave-one-out (Pearson 0.90, median error 0.86 log2).
2. **GTEX seed-family** (fallback). floor0 = no reference anywhere (left un-imputed and flagged, never fabricated — imputing a truly-absent arm would *erase* a real gain).

GTEX collapse-repair repairs the GTEX abundance matrix *at source* before any contribution, so `c_gtex / dHT / acquired_vs_gtex` are collapse-free (the let-7a/ miR-30a/miR-182 collapse artifacts no longer register as fake gainers; `frac_gain_imputable_artifact` = 0/1,410). **Coupling is left on the raw GTEX matrix** (it re-derives its own bundles), so partial-p is untouched. The imputed baseline feeds **only the acquired/dev axis** — never the z-spine or coupling — so all headline coupling findings are baseline-independent.

8.2 Two acquired axes — rank and magnitude

A single rank delta is not enough, because rank **saturates**:

RANK $dHT(m) = rank_tumor(m) - rank_gtex(m)$ (percentile-rank delta on the global pressure budget)
MAGNITUDE $log2fc_tumor_vs_healthy(m) = tcga_tumor_median(m) - healthy_baseline_tcga(m)$ (same C

dHT is cross-platform-robust but an already-abundant arm *cannot gain rank* (max dHT ≈ 0.08 in the 80–95% bin), so **81% of >4-fold-up arms are rank-invisible** — precisely the BC oncomiR surge (miR-182/183/375/93/141, +9 to +13 log2) is *magnitude_only*. The magnitude axis recovers them. **The default acquired readout is both axes:** `acquired_gainer` = rank-gain OR magnitude-gain; 242/721 arms gain on ≥ 1 axis (69 rank-only, 141 magnitude-only, 32 both).

8.3 Trajectory classification, at three resolutions

miRNA (8.3a) on (dHN, dNT, dHT): `stable / progressive {gain,loss} / tumor_acquired / field_effect`
edge (8.3c) join of the miRNA's pressure-move with the per-edge coupling trajectory $\rightarrow$ `joint_edge`

gene (8.3e) acquired_vs_gtex / acquired_vs_nat with the floor0-artifact split as a residual dia

The three legs are dHN = nat-gtex (field effect), dNT = tumor-nat (tumor-specific increment), dHT = tumor-gtex (the **acquired headline**). A **subtype-specificity** secondary class (Tau over PAM50) tags whether an acquired arm is pan or single-subtype (§ glossary; LANDSCAPE_REPORT.md §5). The whole §4.2 acquired story is read off this trajectory — and because it is anchored to the GTEx *state* (never NAT, which sits tumor-ward by field effect), “acquired over healthy” means exactly that.

9. Validation — recovered, deepened, novel

Validation is framed by *what kind of claim* a result is, using the registry tags (S statistical+FDR+adjusted / P pattern / H hypothesis / R reused-known). The three questions are: *did we recover known biology? did we deepen it? did we find anything new?*

9.1 Recovered — the positive controls

The model reproduces textbook biology *without being told to*, which is the evidence it is real:

- **The TS-miRNA roster** falls out of the architecture anti-tumor leaders (miR-30a, miR-24, miR-29b/c, let-7a) and the gene-role oncomiR roster (miR-21/182/10b/93/106b).
- **miR-29→collagen/ECM, miR-17~92→PTEN/replication, let-7/miR-30→mitotic, miR-375 as immune-evasion oncomiR** all emerge as sub-modules and grip-change leaders in the within-pathway NMF — internal validation (if they had *not* appeared the decomposition would be suspect).
- **HE edges couple more than orphan edges** reproduces across cohorts (TCGA and Buffa).
- **Cross-cohort replication (Buffa 2011, 207 independent patients):** Buffa-vs-TCGA partial- ρ concordance +0.32; 67% of TCGA neg-sig edges keep a negative sign — an independent-patient leg the same-patient CPTAC cannot give.

9.2 Deepened — same players, new resolution

The platform’s genuine methodological contribution is **state-resolved and within-program** structure that standard GSEA / single-edge analyses cannot see:

- **State-resolved coupling** (R00 brake-release, 00R acquired-realized): the literature asserts the miRNA→target *pair* and the miRNA’s down-regulation, but rarely measures *the brake being engaged in normal tissue and released in tumor* (strength P/H).
- **Within-program sub-modules:** a Hallmark resolves into a shared pan-subtype backbone
 - named subtype-specialist sub-modules (Basal CD69/ICAM1 immune-restriction in IFN- γ , PTEN-repression localized to a *minority* PI3K/P53 sub-module) — “same known players, newly resolved organization.”
- **Protein-layer confirmation** of the basal miR-17~92→PTEN hub in an independent proteome (§7).
- **CN→regulatory-network attribution** (MH-43): the CN-driven component of miRNA dosage reproduces ~half-to-two-thirds of the bulk repressive network and survives the full aneuploidy+proliferation control stack.

9.3 Novel / candidate-novel — the orphan discovery surface

The cleanest discovery surface is the **orphan edges**: TargetScan/ENCORI-predicted interactions **absent from miRTarBase functional curation** — kept out of the spine precisely so they can serve as an unbiased discovery screen. The funnel:

20,663 orphan edges screened (TargetScan $\cup$ ENCORI, miRTarBase-functional-MTI-absent)
→ 15,512 testable against CPTAC protein
→ 539 protein anti-correlation candidates

→ 182 translational (protein_resid, beyond-mRNA)
→ 93 translational AND TCGA-replicated ← strongest tier (76/93 fully uncured · 29/93 overlapping subsets, not a partition)

Three convergent strengthenings make these more than a prediction list:

- **Method self-validation:** the **miR-29→collagen/ECM axis emerges de novo** (miR-29a→COL11A1 ts_weight 0.80, clipExpNum 15; →COL6A2, →BMP1) — a textbook axis uncured *at these specific targets*. It is the credibility anchor.
- **Triple-cohort validation (MH-38):** of the **492 genuine orphans** (no miRTarBase row at all), **127** were CPTAC-protein-significant, TCGA-replicated, *and* testable in the independent Buffa cohort; of those **108 (85%) keep a negative sign in Buffa** — far above the 50% chance rate — giving **108 triple-cohort-validated orphan edges** (TCGA mRNA · CPTAC protein · Buffa mRNA; 47 reach FDR in Buffa), led by miR-29→ECM (34/108 ECM targets).
- **Composition-robust core (MH-39) — the 17 TargetScan-only orphan edges:** re-testing the prediction-only tier (TS-sequence-only, no CLIP, no curation) after adding PAM50 + ESTIMATE immune/stromal + endothelial + target-CN covariates leaves **17 of 24 survivors holding** (negative beyond-mRNA protein_resid at $q < 0.1$ under the widest covariate set; **11/17** at $q < 0.05$). The 7 drops are exactly the composition artifacts — VCL x2, TPM4, and the hematopoietic miR-142-3p/-5p cluster (WASL/STAM2/RNH1/HOOK3), which load on the CAF/immune axes. The 17 surviving edges, grouped by target (protein ρ = prospective CPTAC, base purity+CIN; resid ρ = protein residualized on its own mRNA under the *widest* covariate stack, with FDR q):

Target (function)	TS-only arm(s) (ts_weight)	protein ρ	resid ρ (q)
PMM1 — mannose metabolism	miR-195-5p (0.27) · miR-16-5p (0.27) · miR-15a-5p (0.30)	−0.40 · −0.39 · −0.39	−0.40 (9e-4) · −0.31 (.012) · −0.30 (.012)
PPM1A — PP2C phosphatase (TGF- β /AKT brake)	miR-19b-3p (0.26) · miR-19a-3p (0.26)	−0.30 · −0.32	−0.28 (.016) · −0.23 (.042)
PURA — transcription / replication	miR-532-5p (0.26) · miR-452-5p (0.36)	−0.45 · −0.33	−0.19 (.093) · −0.18 (.093)
KIF1B — kinesin, pro-apoptotic	miR-15a-5p (0.45)	−0.37	−0.30 (.012)
NCK2 — signalling adaptor	miR-142-3p (0.49)	−0.07	−0.31 (.012)
VCAN — versican (ECM)	miR-107 (0.27)	−0.52	−0.27 (.018)
MAPRE1 — EB1, microtubule +end	miR-429 (0.27)	−0.25	−0.27 (.018)
NUDT4 — Nudix hydrolase	miR-183-5p (0.82)	−0.16	−0.26 (.023)
RAB1A — ER→Golgi traffic	miR-101-3p (0.59)	−0.33	−0.24 (.041)
CERCAM — endothelial adhesion	let-7g-5p (0.34)	−0.35	−0.22 (.057)
SKP1 — SCF ubiquitin-ligase core	miR-452-5p (0.29)	−0.38	−0.21 (.066)
DR1 — NC2 transcriptional repressor	miR-182-5p (0.29)	−0.34	−0.19 (.093)
GLUD1 — glutamate dehydrogenase	miR-30e-5p (0.27)	−0.28	−0.18 (.093)

Two **seed-family convergences** lead and are CAF-neutral / method-independent (they hold even with target-gene CN partialled out): the triple-hit **miR-15/16/195→PMM1** (mRNA-flat, protein-strong = translational) and **miR-19a/b→PPM1A. NCK2** is the instructive single case — its *direct* protein ρ is ≈ 0 yet the **beyond-mRNA residual** is strong (-0.31 , $q=.012$), a translational-layer signal bulk mRNA cannot see. All 17 remain **wet-lab nominations** (a prioritized CLIP/luciferase queue), not validated edges.

What these are and are NOT. They are **wet-lab nominations** (a prioritized CLIP/luciferase queue), not validated edges — protein anti-correlation $\neq$ direct binding, co-expression confound persists (especially ECM $\leftrightarrow$ stroma), and seed-family arms (miR-29a/b/c) cannot be separated. The miR-29 $\rightarrow$ collagen recovery is the anchor; the fully-uncurated dual-prior translational hits and the 17 composition-robust prediction- only edges are the discovery surface.

9.4 The validation ladder, summarized

Rung	Evidence	Example	Strongest tag
Reproduces known	recovers textbook edges unprompted	miR-29 $\rightarrow$ collagen,	R
mRNA coupling	FDR-neg partial- ρ , confounder-adjusted	miR-17~92 $\rightarrow$ PTEN Basal 42/50 Hallmarks	S
State-resolved	brake-release / acquired-realized trajectories	ERBB2/MMP/NOTCH2 R00	P/H
Protein layer	independent CPTAC proteome	PTEN $\leftarrow$ miR-17-5p protein-resid	S
Independent cohort	Buffa replication	+0.32 concordance, 108 triple-validated	S/R
Wet-lab queue	nominations for CLIP/luciferase	17 composition-robust orphans, 76 uncurated	H

10. Cross-cutting design principles and honest limits

Restating the stance (§0.3) as it cashes out, plus the standing caveats (LANDSCAPE_REPORT.md §8):

- **Static weight $\times$ dynamic abundance** keeps “what’s possible” separate from “what this tumor expresses.”
- **Non-circular prior** — edge weights / HE filter / breast re-ranking / gene roles never use TCGA expression, so they cannot leak into the coupling they are tested against. **Checked, not just asserted:** the pressure construction’s candidate ranking reproduces on held-out patients (5-fold CV, train $\rightarrow$ test rank- ρ 1.0), so it is not overfit to the cohort it is validated on (held_out_tuning, MH-47).
- **Two measurements, then their join** — pressure (potential) and coupling (realized) are never conflated; the joint class is the headline.
- **Adjusted and raw, gated and ungated, everywhere** — differences read as sensitivity. The AGO/RISC gate in particular is a *documented sensitivity layer, not a causal model* (it rescales but rarely reorders): capacity is the **co-limiting minimum** of a loadable-AGO term (AGO2-weighted) and a TNRC6/GW182 effector term — both required — and is floored so it can attenuate, never erase (METHODS.md §4).
- **Honest rungs** — potential $\neq$ realization $\neq$ causation; every claim is tagged.

Standing limits, stated plainly:

- **Gene-gene topology cannot see lipid/metabolite-mediated inhibition** (PTEN $\rightarrow$ PIP3) — a true ceiling; the gene-role layer is the partial workaround.

- **De-coupling mechanism is not fully pinned:** CN (0/220) and promoter methylation (1/188) are ruled out, ATAC is mostly null (a 4/134 accessibility-linked minority); the leading remaining candidate is 3'UTR APA/shortening + ceRNA/RBP competition.
- **floor0 arms** (no healthy reference) are a coverage gap, not acquisition — resolve by extending GTEx/miTED coverage, never by imputing absent arms.
- **The prior is pan-context and coarse** (one BC polarity per set); breast refinement is an overlay with no external breast-miRNA cohort yet to validate utility.
- **The pressure construction is one of several near-equivalent choices** — non-overfit out-of-sample but not uniquely optimal: bare un-normalized weighting is marginally ahead on coupling, and the promiscuity denominator is coupling-neutral, earning its place on attribution (dominant-regulator credit + hub de-concentration), not coupling (MH-47).
- **Bulk-RNA / bulk-protein ceiling** — everything is association, not binding; composition (stroma/immune) is controlled but never fully eliminated.

Appendix: terminology glossary

Term	One-line definition	Where
Pressure c / $\text{pressure}(g, s)$	potential repression = expression $\times$ edge weight, summed over a gene's regulators	§4.1
Edge weight $\text{edge}_w(m, g)$	static evidence strength $\div$ promiscuity denominator	§2.3
Contribution $c(m, g, s)$	edge-resolution pressure atom; everything is built from it	§4.1
Acquired / lost pressure	tumor pressure minus a healthy anchor (GTEx or NAT), at edge/gene/hallmark	§4.2, §8
Share (global_abs_share)	a regulator's fraction of a gene's total pressure magnitude (arm-side vs gene-side)	§4.3
Specificity spec / ts_spec	how concentrated an arm's budget is on one gene (focused vs promiscuous hub)	§4.4
Coupling (partial- ρ)	observed, confounder-adjusted anti-correlation = <i>realized</i> pressure	§6
Module composite	partial- ρ of an arm vs the mean- z of its whole target set (coordinated repression)	§6.3
Confounders	CPE (purity), HRD, target CN, proliferation (3 proxies), ESTIMATE+epi	§6.2
Joint edge class	the pressure $\times$ coupling join (acquired_realized, lost/R00, constitutive, ...)	§6.4
Architecture	topology (output_influence) + tumor-direction prior + gene-role malignancy on a set	§5

Term	One-line definition	Where
pro_tumor_pressure	prior × propagated program-output change (brake-release vs engine-damage)	§5.2
mal_pro_tumor	gene-role driver call: –malignancy_sign × pressure (curated; +DepMap continuous)	§5.3–5.4
L2 / L1 / L1b	protein-direct / RNA-protein gap / protein-residualized-on-mRNA (beyond-mRNA)	§7.1
dHT / log2fc	acquired rank axis (saturating) / acquired magnitude axis (recovers oncomiR surge)	§8.2
Orphan edge	TargetScan/ENCORI-predicted, miRTarBase-uncurated — the discovery screen	§9.3
AGO/RISC gate	per-sample bounded multiplier; capacity = min(AGO2-weighted load, TNRC6 effector) — co-limited, floored, sensitivity layer not mechanism	METHODS.md §4
NMF program	unsupervised pressure factor (subtype-divergent programs vs shared acquired backbone)	LANDSCAPE_REPORT.md §6
Tau (subtype specificity)	Yanai index of how single-subtype an arm's acquired pressure is	LANDSCAPE_REPORT.md §5
Strength tags	S statistical / P pattern / H hypothesis / R reused-known	§9

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